

Clique and chromatic numbers of binary subspace orthogonality graphs

Haobo Ma*

Reza Nikandish†

Wenlin Zhang‡

October 8, 2026

Extended working draft

Abstract

Let $\mathcal{O}_n^*$ be the graph whose vertices are the nonzero subspaces of $\mathbb{F}_2^n$, with two distinct vertices adjacent when every vector of one is orthogonal to every vector of the other, and let Γ_n be its induced subgraph on the one-dimensional subspaces. We prove

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\} \quad (n \geq 1),$$

where $N(r) = \sum_{d=1}^r \binom{r}{d}_2$ is the number of nonzero subspaces of $\mathbb{F}_2^r$. The proof is algebraic: it sums the radicals of an arbitrary clique and passes to the resulting orthogonal quotient. In dimension six we show, with computer-assisted certificates, that $\chi(\mathcal{O}_6^*) = 15$ and $\chi(\Gamma_6) = 12$, so the full subspace graph needs exactly three more colors than its line subgraph. The lower bound for Γ_6 comes from a weighted independent-set bound; both upper bounds are explicit colorings verified by independent checkers, and the clique theorem and the 15-coloring are also formalized in Lean, with native evaluation for the finite checks in the latter. We further show that no proper coloring of $\mathcal{O}_6^*$ with the prescribed clique colors is invariant under the stabilizer of the fixed totally isotropic 3-space, and we prove $\chi(\Gamma_7) \geq 19$ and hence $\chi(\mathcal{O}_7^*) \geq 19$, exceeding the clique number 16. The dimension-seven obstruction combines quadratic parity with a universal completion theorem for eight disjoint Lagrangians. We also derive common count and charge constraints for a possible nineteen-coloring. The exact dimension-seven chromatic numbers remain open.

2020 Mathematics Subject Classification: 05C15, 05C25, 05C69, 13A15, 68V20

Key Words: Orthogonality graph, binary subspace, clique number, chromatic number, formal verification.

1 Introduction

Let $V_n = \mathbb{F}_2^n$ denote the n -dimensional vector space over the field with two elements, equipped with the standard nondegenerate bilinear form

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$$

*ChronoAI Pte Ltd; The Omega Institute. auric@aelf.io.

†Department of Mathematics, Jundi-Shapur University of Technology, Dezful, Iran. r.nikandish@ipm.ir (corresponding author).

‡National University of Singapore; The Omega Institute. e1327962@u.nus.edu.

The *orthogonality graph* $\mathcal{O}_n^*$ has as its vertices the nonzero linear subspaces of V_n . Two distinct vertices U, W are adjacent when every vector of U is orthogonal to every vector of W , that is, $U \subseteq W^\perp$. Its induced subgraph on the one-dimensional subspaces is denoted by Γ_n . Since each line of V_n contains a unique nonzero vector, Γ_n can equivalently be described as the graph on the $2^n - 1$ nonzero vectors of V_n with two distinct vectors adjacent when their dot product is zero.

The graph $\mathcal{O}_n^*$ belongs to a natural family of graphs arising from finite classical polar spaces. Its line subgraph Γ_n is a finite-field analogue of the Kneser graph: two lines are adjacent when their representative vectors are orthogonal, rather than when their supports are disjoint. The chromatic number of the Kneser graph was determined by Lovász [10] using topological methods; the clique and chromatic numbers of polarity graphs and their relatives have been studied extensively in finite geometry and algebraic graph theory [6, 9, 7]. The subspace graph $\mathcal{O}_n^*$ refines the line graph by admitting higher-dimensional subspaces as additional vertices, and it is this enlargement that makes the chromatic problem substantially different.

Motivation and background

The graph $\mathcal{O}_n^*$ arises naturally in two independent contexts. On the combinatorial side, it is the graph of pairwise orthogonal subspaces of a finite vector space, a natural object in the theory of polar spaces [8]. On the algebraic side, it is isomorphic to the annihilating-ideal graph of a family of finite local rings introduced by Nikandish [13], and thus provides a bridge between the theory of graphs over finite fields and the study of graphs associated with commutative rings [3, 1, 2, 4, 5].

The bridge is as follows. For

$$R_n = \frac{\mathbb{F}_2[x_1, \dots, x_n, z]}{(z^2, zx_i, x_i x_j (i \neq j), x_i^2 - z)},$$

the map $U \mapsto (z) + U$ is a bijection between the subspaces of V_n and the nonzero proper ideals of R_n , and $I_U I_W = 0$ if and only if $U \perp W$ [13]. The zero subspace corresponds to the ideal (z) , which is a universal vertex, so

$$\chi(\mathbb{A}\mathbb{G}(R_n)) = \chi(\mathcal{O}_n^*) + 1, \quad \omega(\mathbb{A}\mathbb{G}(R_n)) = \omega(\mathcal{O}_n^*) + 1.$$

In particular, any statement about $\mathcal{O}_n^*$ translates directly into a statement about $\mathbb{A}\mathbb{G}(R_n)$, and conversely.

Problem 1.1. Determine the exact value of $\omega(\mathcal{O}_n^*)$ for every n .

Problem 1.2. Determine $\chi(\mathcal{O}_n^*)$ for $n \geq 6$ and investigate the relation between $\chi(\mathcal{O}_n^*)$ and $\omega(\mathcal{O}_n^*)$.

Problem 1.3. Determine whether there exist infinitely many $n \geq 1$ with $\chi(\mathcal{O}_n^*) > \omega(\mathcal{O}_n^*)$.

Results

Put

$$N(r) = \sum_{d=1}^r \begin{bmatrix} r \\ d \end{bmatrix}_2,$$

where $\begin{bmatrix} r \\ d \end{bmatrix}_2$ is the Gaussian binomial coefficient, and set $N(0) = 0$. Our results are

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\} \quad (n \geq 1), \quad (1)$$

and

$$\chi(\mathcal{O}_6^*) = 15, \quad \chi(\Gamma_6) = 12. \quad (2)$$

In dimension seven we prove

$$\chi(\Gamma_7) \geq 19, \quad \chi(\mathcal{O}_7^*) \geq 19, \quad (3)$$

whereas (1) gives $\omega(\mathcal{O}_7^*) = 16$. The first formula answers Problem 1.1 in all dimensions. The second answers the dimension-six case of Problem 1.2 and quantifies the separation between coloring the line graph and coloring the full subspace graph: in dimension six, passing from lines to all nonzero subspaces increases the chromatic number by exactly three.

The clique argument is algebraic: it sums the radicals of an arbitrary clique and passes to the resulting orthogonal quotient. The full-graph chromatic upper bound uses an explicit finite certificate, while the line-graph equality follows from a weighted independent-set bound and a checked 12-coloring. For dimension seven, a sharp capacity bound on line color classes reduces a hypothetical eighteen-coloring to two saturated patterns. Quadratic parity excludes one, and a Lagrangian completion and hole-capacity argument excludes the other. This proof is independent of the subsequent analysis of possible nineteen-colorings. The verification details and the scope of the Lean formalization are collected in Section 8.

Table 1 records the known small-dimensional values. The full subspace graphs satisfy $\chi > \omega$ for $n = 4, 5$, whereas $\chi(\mathcal{O}_6^*) = \omega(\mathcal{O}_6^*) = 15$. In dimension six, the three-color separation in (2) is between the full subspace graph and its line subgraph.

n	$\omega(\mathcal{O}_n^*)$	$\chi(\mathcal{O}_n^*)$
1	1	1
2	2	2
3	3	3
4	4	5
5	5	8
6	15	15
7	16	≥ 19

Table 1: Clique and chromatic numbers of $\mathcal{O}_n^*$ for small n . The values for $n \leq 5$ were computed in [13]; the value $\chi(\mathcal{O}_6^*) = 15$ and the bound $\chi(\mathcal{O}_7^*) \geq 19$ are established here. For general n , the clique number is given by (1); the chromatic number for $n \geq 7$ remains open.

The paper is organized as follows. Section 2 contains the proof of (1). Section 3 treats the line graph Γ_6 and establishes $\chi(\Gamma_6) = 12$. Section 4 proves $\chi(\mathcal{O}_6^*) = 15$. Section 5 describes the geometry of the precoloring and records the obstructions to a structural formula. Section 6 proves (3). Section 7 derives necessary nineteen-color constraints, gives a dimension reduction, and states the remaining open problems. Section 8 collects the computational certificates and the Lean formalization.

2 The exact clique number

Let $N(r)$ denote the number of nonzero subspaces of $\mathbb{F}_2^r$, with $N(0) = 0$. Equivalently, $N(r) = \sum_{d=1}^r \binom{r}{d}_2$.

Theorem 2.1. *For every positive integer n ,*

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\}.$$

Proof. We prove the upper and lower bounds separately.

Lower bound. The n coordinate lines $\langle e_1 \rangle, \dots, \langle e_n \rangle$ are pairwise orthogonal and form a clique of size n . For the second term, let $r = \lfloor n/2 \rfloor$ and take

$$T = \text{span}\{e_1 + e_2, e_3 + e_4, \dots, e_{2r-1} + e_{2r}\}.$$

The form $\langle \cdot, \cdot \rangle$ vanishes on T , so every nonzero subspace of T is orthogonal to every other one. Hence the nonzero subspaces of T form a clique of size $N(r)$. When n is odd, $\dim T^\perp = r + 1$, so $T^\perp$ is a distinct additional vertex orthogonal to every subspace of T . These two constructions attain both terms of the maximum. They extend the coordinate and isotropic constructions in [13, Sections 3–4].

Upper bound. Let $\mathcal{C}$ be a clique in $\mathcal{O}_n^*$. For each $U \in \mathcal{C}$ put

$$\text{rad}(U) = U \cap U^\perp,$$

and define

$$R = \sum_{U \in \mathcal{C}} \text{rad}(U), \quad r = \dim R.$$

By construction, $R \subseteq U^\perp$ for every $U \in \mathcal{C}$, and the summands are pairwise orthogonal. Hence $R \subseteq R^\perp$, so R is totally isotropic and $2r \leq n$. Moreover,

$$U \subseteq R^\perp, \quad R \subseteq U^\perp, \quad U \cap R = \text{rad}(U) \quad (U \in \mathcal{C}).$$

For the last equality, one inclusion is the definition of R ; the other follows from $R \subseteq U^\perp$.

Since the standard dot form is nondegenerate, $\dim(R^\perp/R) = n - 2r$. The form descends to the quotient $Q = R^\perp/R$. Write $\bar{U}$ for the image of U in Q . We claim that the restricted form on $\bar{U}$ is nondegenerate. Indeed, if $u \in U$ represents a vector annihilating $\bar{U}$, then $u \in \text{rad}(U) \subseteq R$, so its image in Q is zero.

The spaces $\bar{U}$ are pairwise orthogonal in Q . Their nonzero members form an indexed direct sum: in any relation $\sum_U x_U = 0$ with $x_U \in \bar{U}$, pairing with arbitrary vectors of one $\bar{W}$ and using the nondegeneracy of the restriction to $\bar{W}$ gives $x_W = 0$. Hence there are at most $\dim Q = n - 2r$ nonzero images.

The members of $\mathcal{C}$ with zero image are precisely those contained in R , and there are at most $N(r)$ of them. Therefore

$$|\mathcal{C}| \leq N(r) + n - 2r. \quad (4)$$

For $r = 0$, this gives $|\mathcal{C}| \leq n$. For every $s \geq 1$, embedding $\mathbb{F}_2^s$ as a hyperplane in $\mathbb{F}_2^{s+1}$ gives

$$N(s+1) \geq N(s) + 2,$$

because an outside line and the whole ambient space are two distinct additional subspaces. Hence $N(s) - 2s$ is nondecreasing for $s \geq 1$. If $r \geq 1$, putting $t = \lfloor n/2 \rfloor$ in (4) gives

$$|\mathcal{C}| \leq N(t) + n - 2t = N(t) + (n \bmod 2).$$

This completes the proof of the upper bound and hence of the theorem. $\square$

Corollary 2.2. For $n = 6$,

$$\omega(\mathcal{O}_6^*) = N(3) = 15.$$

An explicit clique of size 15 is given by the nonzero subspaces of $T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\}$, which is a totally isotropic 3-space.

3 The line graph in dimension six

This section determines the chromatic number of the line graph Γ_6 , whose vertices are the 63 nonzero vectors of $\mathbb{F}_2^6$ with two distinct vectors adjacent when their dot product vanishes. We proceed in two steps. First, we bound the size of any independent set in Γ_6 by a weighted count on even- and odd-weight vectors; this yields the lower bound $\chi(\Gamma_6) \geq 12$. Second, we exhibit an explicit 12-coloring of the 63 nonzero vectors, verified by an independent checker, which gives the matching upper bound. Together these prove $\chi(\Gamma_6) = 12$. The lower bound is the substantial combinatorial input; the upper bound is a finite certificate recorded in Section 8.

Lemma 3.1. *An independent set in Γ_6 containing t even-weight vectors and o odd-weight vectors satisfies $3t + 4o \leq 19$. Its cardinality is at most six; cardinality six requires $(t, o) = (5, 1)$.*

Proof. Write $j = (1, 1, 1, 1, 1, 1)$. The even-weight vectors lie in $j^\perp$, whose restricted form has radical $\text{span}(j)$ and rank four. If $o = 0$, the Gram matrix of the selected vectors is $J_t + I_t$, of rank t for even t and $t - 1$ for odd t . Thus $t \leq 5$.

If $o > 0$, fix an odd selected vector a . For each even selected vector e , the vector $z_e = e + a$ belongs to $a^\perp$. These vectors are orthonormal and span a nondegenerate t -space A . The ambient space $a^\perp$ is nondegenerate of dimension five. The span D of the vectors $u + a$, over odd selected vectors u , is totally isotropic and orthogonal to A inside $a^\perp$. Hence $2 \dim D \leq 5 - t$. The odd selected vectors are distinct members of $a + D$, so $o \leq 2^{\dim D}$. Consequently $o \leq 4$ for $t = 0, 1$, $o \leq 2$ for $t = 2, 3$, and $o \leq 1$ for $t = 4, 5$. These bounds prove both claims. $\square$

Theorem 3.2. $\chi(\Gamma_6) = 12$.

Proof. There are 31 nonzero even-weight and 32 odd-weight vectors. Their total weight, using weights three and four respectively, is 221. By Lemma 3.1, every color class has weight at most 19, giving $\chi(\Gamma_6) \geq \lceil 221/19 \rceil = 12$. For the upper bound, interpret an integer x as the vector whose coordinate i is its i th binary digit. The certificate `results/line-coloring-12.json` partitions the 63 nonzero vectors into 12 classes. The independent standard-library checker `develop/check_line_chromatic_exact.py` checks all 1953 pairs and verifies that each of the 961 orthogonality edges has distinct colors. $\square$

For completeness, the twelve color classes, in the same binary integer coordinates, are displayed below. They partition $\{1, \dots, 63\}$, and their within-class dot products can be checked directly.

$$\begin{aligned} &\{8, 13, 29, 42, 47\}, \quad \{2, 26, 38, 62, 63\}, \quad \{20, 37, 39, 45, 48\}, \\ &\{6, 11, 18, 34, 53, 60\}, \quad \{15, 30, 46, 49, 59\}, \quad \{32, 40, 50, 51, 54\}, \\ &\{14, 19, 24, 41, 52\}, \quad \{12, 27, 43, 55, 57, 58\}, \quad \{4, 7, 28, 31, 36\}, \\ &\{16, 21, 23, 56, 61\}, \quad \{10, 22, 25, 35, 44\}, \quad \{1, 3, 5, 9, 17, 33\}. \end{aligned}$$

Any 12-coloring has a class of size six, since $12 \cdot 5 < 63$. By the lemma it consists of an odd vector a and five even vectors e . The vectors a and $e + a$ form an orthonormal basis. An isometry therefore maps this class to $\{1, 3, 5, 9, 17, 33\}$. Fixing this class reduces the remaining question to an 11-coloring of the induced graph on the other 57 vectors. The direct Boolean encoding uses 627 variables and 11 772 clauses: each vertex has exactly one of eleven colors, and each of the 780 edges forbids a shared color. The solver supplies a satisfying assignment; adjoining the fixed class gives the displayed certificate. The proof of the upper bound depends on the independent certificate check, rather than on the solver's correctness.

4 The full chromatic number in dimension six

This section establishes the main chromatic result of the paper, $\chi(\mathcal{O}_6^*) = 15$. The lower bound is immediate: the fifteen nonzero subspaces of the fixed totally isotropic 3-space $T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\}$ are pairwise orthogonal and form a 15-clique, so $\chi(\mathcal{O}_6^*) \geq 15$. The upper bound is the substantial direction: we produce an explicit 15-coloring of all 2824 nonzero subspaces and verify it against every orthogonality edge. The certificate is recorded in the accompanying data file and checked independently by the checker of Section 8. The closing part of this section records the list-size structure that guides both the search for the certificate and the geometric analysis of Section 5.

Theorem 4.1. $\chi(\mathcal{O}_6^*) = 15$.

Proof. Let $T = \text{span}(3, 12, 48)$ in binary integer coordinates, with bit $i - 1$ representing the coefficient of e_i . Thus its generators are $e_1 + e_2, e_3 + e_4, e_5 + e_6$. The form vanishes on T , so its seven lines, seven planes and T itself form a 15-clique. Hence $\chi(\mathcal{O}_6^*) \geq 15$.

For the upper bound, the file `results/full-15-coloring.json` gives a basis and a color in $\{0, \dots, 14\}$ for every nonzero subspace. The independent checker reconstructs each span and verifies basis independence, distinctness, and the dimension counts

$$63, 651, 1395, 651, 63, 1.$$

These equal the Gaussian binomial coefficients for $d = 1, \dots, 6$, which proves completeness of the catalogue. For all 3 986 076 pairs of distinct entries the checker tests orthogonality on the basis vectors. Bilinearity identifies this test with graph adjacency. All 44 968 orthogonality edges have distinct endpoint colors. This establishes the upper bound $\chi(\mathcal{O}_6^*) \leq 15$, and therefore equality. $\square$

Remark 4.2. To find the certificate, fix distinct colors on the 15 nonzero subspaces W of T . Every 15-coloring uses all colors on this clique, so relabeling colors makes this precoloring an equivalent instance. The forbidden colors for a vertex $U \not\subseteq T$ are exactly those $W \subseteq H(U) = U^\perp \cap T$. If $H(U) = T$, then $U \subseteq T^\perp = T$, a contradiction. Thus $\dim H(U) \in \{0, 1, 2\}$, giving lists of sizes

$$15 - N(\dim H(U)) \in \{15, 14, 11\}.$$

Remark 4.3. A vertex whose residual degree is smaller than its list size can be deleted and restored greedily after the remaining graph is colored. Repeated deletion removes 729 uncolored vertices and leaves 2 080 vertices with 39 400 edges. The list-coloring encoding has 28 600 variables and 627 510 clauses. There is one Boolean variable for each allowed vertex-color pair. Each vertex has an at-least-one clause and pairwise at-most-one clauses; each edge forbids simultaneous use of any color common to its endpoints' lists. Hence satisfying assignments are exactly the proper list colorings of the residual graph. A satisfying assignment was extended in reverse deletion order to produce the full certificate checked above.

5 Geometry of the precoloring

In this section we describe the geometric color lists and profiles of the 15-coloring constructed in Section 4. Keep the notation of that section: T is the totally isotropic 3-space

$$T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\},$$

$\mathcal{C}$ is the 15-clique formed by the nonzero subspaces of T , and each $W \in \mathcal{C}$ carries its own color c_W .

For a vertex U of $\mathcal{O}_6^*$ outside $\mathcal{C}$, define

$$H(U) = U^\perp \cap T, \quad K(U) = U \cap T.$$

Since $T^\perp = T$ and $U \not\subseteq T$, we have $H(U) \neq T$, so $\dim H(U) \in \{0, 1, 2\}$. The following observation is the key to the precoloring.

Lemma 5.1. *Let U be a vertex of $\mathcal{O}_6^*$ outside $\mathcal{C}$ and let $W \in \mathcal{C}$. Then U is adjacent to W if and only if $W \subseteq H(U)$. Consequently, the allowed color list of U is*

$$L(U) = \{c_W : 0 \neq W \leq T, W \not\subseteq H(U)\},$$

and its size is

$$|L(U)| = \begin{cases} 15, & \dim H(U) = 0, \\ 14, & \dim H(U) = 1, \\ 11, & \dim H(U) = 2. \end{cases}$$

Proof. By definition, U is adjacent to W if and only if $W \subseteq U^\perp$. Since $W \subseteq T$, this is equivalent to $W \subseteq U^\perp \cap T = H(U)$. The list size is $15 - N(\dim H(U))$, where $N(0) = 0$, $N(1) = 1$, and $N(2) = 4$; this gives 15, 14, and 11 in the three cases. $\square$

Table 2 groups the $2809 = 2824 - 15$ vertices outside $\mathcal{C}$ by the dimensions of U , $H(U)$, and $K(U)$.

$\dim U$	$\dim H(U)$	$\dim K(U)$	Vertices
1	2	0	56
2	1	0	448
2	2	1	196
3	0	0	512
3	1	1	784
3	2	2	98
4	0	1	448
4	1	2	196
4	2	3	7
5	0	2	56
5	1	3	7
6	0	3	1

Table 2: Distribution of the 2809 vertices outside the 15-clique, according to the triple $(\dim U, \dim H(U), \dim K(U))$.

The table has 12 dimension profiles. Its 357 vertices with $\dim H(U) = 2$ are the most constrained, having 11 allowed colors. These lists describe the constraints imposed by the clique. The full certificate in Section 4 establishes that the precoloring extends by checking every orthogonality edge, including those between vertices outside the clique.

5.1 Towards a structural description

Retaining the actual subspaces $H(U)$ and $K(U)$, rather than their dimensions alone, gives the exact profiles $(\dim U, H(U), K(U))$. Grouping the archived coloring by this triple produces 240 profiles

outside the clique; 139 receive more than one color and the remaining 101 receive a single color. Thus even the exact profile does not determine the color in this fixed certificate. These statistics are recorded in `results/coloring-geometry.json` and reproduced by `develop/analyze_geometry.py`. The statistics alone do not settle whether a different proper coloring could depend on these profiles. The following obstruction rules out such a rule even with the quotient-form isometry type added.

Proposition 5.2. *No proper coloring of $\mathcal{O}_6^*$ depends solely on $(\dim U, H(U), K(U))$ and the isometry type of the restricted nondegenerate bilinear form on $U/(U \cap U^\perp)$.*

Proof. The distinct lines $U = \text{span}(e_1)$ and $V = \text{span}(e_2)$ are adjacent. Both have dimension one, $H(U) = H(V) = \text{span}(e_3 + e_4, e_5 + e_6)$, and $K(U) = K(V) = \{0\}$. Their restricted forms have Gram matrix $[1]$, so their radicals are zero and their quotients are isometric. A rule depending only on the stated data assigns equal colors to these adjacent vertices, a contradiction. $\square$

The computation in `results/geometry-profile-audit.json` retains the exact subspaces: adding quotient dimension gives 263 profiles, 160 multicolored; adding alternating status makes no further split. None of the original 139 multicolored profiles is completely resolved. These are finite statistics of the fixed witness, distinct from Proposition 5.2.

Let G be the stabilizer of T in the orthogonal group of V_6 , acting on clique color labels by $g \cdot c_W = c_{gW}$.

Corollary 5.3. *No proper 15-coloring with the prescribed clique colors satisfies $c(gU) = g \cdot c(U)$ for every $g \in G$.*

Proof. The coordinate transposition $(1, 2)$ is orthogonal and fixes every vector of T , hence every clique subspace and every color label. It exchanges the adjacent lines $\text{span}(e_1)$ and $\text{span}(e_2)$. Equivariance would give these lines the same color, a contradiction. $\square$

5.2 Auxiliary choices and symmetry breaking

Proposition 5.2 and Corollary 5.3 show that a proper coloring cannot depend solely on the specified exact profile and quotient isometry type, and that a proper 15-coloring cannot be equivariant under the full stabilizer of T with its induced clique-label action. A structural description must distinguish vertices identified by these data and break the obstructing symmetry. Choosing auxiliary data is one way to do so.

A concrete example is a complement S to T in V_6 , so that $V_6 = T \oplus S$. The choice

$$S = \text{span}\{e_1, e_3, e_5\}$$

is a complement because the generators of T together with those of S form a basis of V_6 . It distinguishes the adjacent lines $\text{span}(e_1)$ and $\text{span}(e_2)$, since

$$\dim(\text{span}(e_1) \cap S) = 1 \quad \text{while} \quad \dim(\text{span}(e_2) \cap S) = 0.$$

Thus this fixed auxiliary choice breaks the coordinate swap responsible for Corollary 5.3.

A direct check on the certificate shows that the two adjacent lines receive distinct colors, 10 for $\text{span}(e_1)$ and 9 for $\text{span}(e_2)$, while every geometric quantity printed by the diagnostic `develop/file_separator.py` is identical. The output and the complement check are recorded in `results/adjacent-line-diagnostic.json`. The auxiliary datum $\dim(U \cap S)$ separates this pair; whether it explains further color choices in the certificate is a separate question.

Whether such complement-based data support a complete structural 15-coloring remains open, as a possible approach to Problem 7.2 in Section 7.

Intersection with S must be distinguished from projection onto S . Let π_S be projection along T , for any complement S .

Proposition 5.4. *The pair $(U \cap T, \pi_S(U))$ and the exact profile $(\dim U, H(U), K(U))$ determine each other. No proper coloring of $\mathcal{O}_6^*$ depends solely on this pair.*

Proof. Write $P = \pi_S(U)$ and $K = U \cap T$. Since the form vanishes on T , $H(U)$ is the annihilator of P in T . The pairing between T and S is perfect: a vector of T orthogonal to S is orthogonal to all of $V_6 = T \oplus S$, hence zero, and both summands have dimension three. Thus P is the annihilator of $H(U)$ in S . Rank-nullity gives $\dim U = \dim K + \dim P$, proving the equivalence. The coloring obstruction follows from Proposition 5.2. $\square$

For the displayed complement, $\pi_S(e_1) = \pi_S(e_2) = e_1$, so the projection pair fails to distinguish those adjacent lines. The computation in `results/complement-projection.json` gives the same 240 exact groups, 139 multicolored, as before; Proposition 5.4 explains this identity. For fixed K and P , the remaining placement is encoded by a linear map $f : P \rightarrow T/K$: if $t + s \in U$, set $f(s) = t + K$. Then $U = \{t + s : s \in P, t + K = f(s)\}$ and $U \cap S = \ker f$. Extracting a structural 15-color rule from these lift data remains open.

Proposition 5.5. *For the displayed T and S , no proper coloring of $\mathcal{O}_6^*$ depends solely on K , P , and the rank, image and kernel of f .*

Proof. In binary integer coordinates take $U = \text{span}(13, 6)$ and $W = \text{span}(9, 14)$. Their distinct spans are $\{0, 6, 11, 13\}$ and $\{0, 7, 9, 14\}$; all four cross dot products of the given bases are zero. Both planes have $K = \{0\}$, $P = \text{span}(1, 4)$, rank two, image $\text{span}(3, 12)$ and kernel zero. On the basis $(1, 4)$ their lift-map values are respectively $(12, 15)$ and $(15, 3)$. A rule using only the common data assigns equal colors to these adjacent planes. $\square$

The finite refinement retaining K, P and these features gives 1662 groups, 160 multicolored. Of the original 139 multicolored groups, 87 split entirely into single-color groups, while 52 remain unresolved; the exact comparison is recorded in `results/lift-map-matrix-check.json`. These statistics describe the fixed witness; Proposition 5.5 excludes every proper coloring solely from the stated data.

For fixed K, P , the complete values of f on a basis $s_1, \dots, s_p$ determine U . Choose representatives $t_i \in T$ of $f(s_i)$; then $U = K + \text{span}(t_1 + s_1, \dots, t_p + s_p)$, since subtracting a lift of the projection of any $u \in U$ leaves an element of K . Changing representatives by elements of K does not change this span. The check in `results/lift-map-values-check.json` reconstructs all 2809 non-clique vertices from these data, giving singleton profiles. Value lists without their domain and quotient data do not suffice: $\text{span}(e_1)$ and $\text{span}(e_3)$ are adjacent and both have the one-entry zero list, but different projection domains. The full representation thus recovers vertices; a structural assignment of 15 colors remains to be found.

The following obstruction gives an explicit pair of adjacent planes in one orbit of the stabilizer of T , ruling out every coloring invariant under that stabilizer. It uses the geometry of the coordinate splitting $V_6 = T \oplus S$ with $S = \text{span}\{e_1, e_3, e_5\}$.

Proposition 5.6 (Stab(T)-orbit obstruction). *Let $G = \text{Stab}_{O_6(\mathbb{F}_2)}(T)$ be the stabilizer of T in the orthogonal group. The distinct orthogonal planes $U = \text{span}(13, 6)$ and $W = \text{span}(9, 14)$ lie in the same G -orbit. Consequently, no proper 15-coloring of $\mathcal{O}_6^*$ with the prescribed clique colors is G -equivariant.*

Proof. With $T = \text{span}(3, 12, 48)$ and $S = \text{span}(1, 4, 16)$, the self-adjoint map $B : S \rightarrow T$ given by $B(1) = 3$, $B(4) = 12$, $B(16) = 0$ has the symmetric matrix $\text{diag}(1, 1, 0)$ in the paired bases $(1, 4, 16)$ and $(3, 12, 48)$. Since T is totally isotropic and $\langle Bs, s' \rangle = \langle s, Bs' \rangle$, the two cross terms cancel in characteristic two. Thus the shear $g_B(t + s) = t + B(s) + s$ preserves the bilinear form. It is an involution fixing T pointwise, so it belongs to G . Direct computation gives

$$g_B(13) = 14, \quad g_B(6) = 9,$$

so $g_B(U) = \text{span}(14, 9) = \text{span}(9, 14) = W$. Since U and W are distinct and orthogonal, the claim follows. $\square$

The shear fixes every clique label, so Proposition 5.6 supplies a plane witness for the equivariance obstruction of Corollary 5.3. It also rules out invariant colorings with any number of labels: vertices in one G -orbit would receive the same color. The two planes connect this orbit obstruction directly to the lift-map analysis of Proposition 5.5.

Two further directions remain for a structural description.

1. *Enrichment of the invariants.* To recover the current coloring, an additional invariant must distinguish vertices within the multi-color exact profiles. Proposition 5.2 shows that the isometry type of the nondegenerate quotient is still insufficient when combined with the exact profile. A finer rule must distinguish the placement of vertices such as the two lines in that proof. The subspaces $H(U)$ and $K(U)$ have a relative position in T , but neither is assumed to contain the other. For example, in binary coordinates $U = \text{span}(1, 3)$ gives $K(U) = \text{span}(3)$ and $H(U) = \text{span}(12, 48)$.
2. *Restricted symmetry or auxiliary choices.* Corollary 5.3 rules out equivariance under the full stabilizer, even when clique color labels are permuted. A structural construction must break some of this symmetry, for example by choosing auxiliary coordinate data such as the complement S above, or restricting to a smaller subgroup. No such construction is asserted here.

6 A nineteen-color lower bound in dimension seven

The clique formula gives $\omega(\mathcal{O}_7^*) = 16$. We prove that its line subgraph already requires at least nineteen colors. Put $z = (1, \dots, 1) \in \mathbb{F}_2^7$ and $E = z^\perp$. Since $z \cdot z = 1$, the restriction of the dot product to E is a nondegenerate alternating form on a six-dimensional space. We call vectors in E even and those outside E odd. A Lagrangian in E is a three-dimensional totally isotropic subspace.

Lemma 6.1. *An independent set in Γ_7 avoiding z has at most seven vertices. An independent set with eight vertices is $z + L$ for a Lagrangian $L \leq E$. An independent set of size seven avoiding z has parity counts $(7, 0)$, $(6, 1)$ or $(0, 7)$.*

Proof. Distinct vectors in an independent set have dot product one. Let m be its number of even vectors and o its number of odd vectors. If $o = 0$, the Gram matrix is $J_m + I_m$. Its rank is m for even m and $m - 1$ for odd m , so $m \leq 7$.

If $o > 0$, fix a selected odd vector a . For each selected even vector u , put $b_u = u + a$. These vectors form an orthonormal set in the nondegenerate six-space $a^\perp$ and span a nondegenerate m -space B . The span D of $w + a$, over selected odd vectors w , is totally isotropic in $B^\perp \cap a^\perp$. Hence $2 \dim D \leq 6 - m$, and the selected odd vectors lie in $a + D$. Thus

$$o \leq 2^{\lfloor (6-m)/2 \rfloor}.$$

For $m = 1, \dots, 6$ the bounds on o are 4, 4, 2, 2, 1, 1, respectively. A class containing an even vector therefore has at most seven members, and size seven requires $(m, o) = (6, 1)$.

If $m = 0$, write each member as $z + e$ with $e \in E$. The selected e are pairwise orthogonal, so their span is isotropic and has dimension at most three. Eight distinct projections must fill a Lagrangian, including zero; their class is $z + L$ and contains z . If z is excluded, there are at most seven members. Seven such projections fill $L \setminus \{0\}$ for a Lagrangian L . $\square$

The following completion lemma holds for every eight-member partial spread; no classification of partial spreads is needed.

Lemma 6.2. *Let $L_1, \dots, L_8$ be Lagrangians in E with pairwise intersection $\{0\}$. The seven nonzero points outside their union are $M \setminus \{0\}$ for a uniquely determined Lagrangian M .*

Proof. For nonzero $e \in E$, the hyperplane $e^\perp$ has 31 nonzero points. Its intersection with L_i has seven nonzero points if $e \in L_i$ and three otherwise, since $L_i = L_i^\perp$. If e is covered, $e^\perp$ contains $7 + 7 \cdot 3 = 28$ covered points and three holes. If e is a hole, it contains $8 \cdot 3 = 24$ covered points and all seven holes. In particular, the holes are pairwise orthogonal. Their span is isotropic and has dimension at most three; seven distinct nonzero points force dimension three. This span is the required M . $\square$

Theorem 6.3. $\chi(\Gamma_7) \geq 19$ and $\chi(\mathcal{O}_7^*) \geq 19$. In particular, $\chi(\mathcal{O}_7^*) - \omega(\mathcal{O}_7^*) \geq 3$.

Proof. Deleting z leaves 126 vertices. Lemma 6.1 gives $\chi(\Gamma_7 - z) \geq 18$. Suppose that Γ_7 has an eighteen-coloring. Since $18 \cdot 7 < 127$, one class has eight members. It is $z + T$ for a Lagrangian T , and all other seventeen classes have size seven. If A, B, C count their pure even, mixed $(6, 1)$ and pure odd classes, respectively, then

$$7A + 6B = 63, \quad B + 7C = 56, \quad A + B + C = 17.$$

The nonnegative solutions are $(A, B, C) = (9, 0, 8)$ and $(3, 7, 7)$.

To exclude the first, choose a symplectic basis of E and write $q(x, y) = x \cdot y$ for $x, y \in \mathbb{F}_2^3$. This quadratic refinement takes the value one at $7 \cdot 4 = 28$ points, so its sum over all nonzero points of E is zero in $\mathbb{F}_2$. For an even heptad $H = \{u_1, \dots, u_7\}$, its Gram matrix $J_7 + I_7$ has rank six and kernel spanned by the all-ones vector. The seven vectors in E have a nonzero linear relation; pairing it with the u_i shows that $\sum_i u_i = 0$. Polarization gives

$$0 = q\left(\sum_i u_i\right) = \sum_i q(u_i) + \binom{7}{2},$$

so each even heptad has odd quadratic sum. Nine disjoint heptads cannot partition all 63 nonzero points of E , whose total quadratic sum is even. This excludes $(9, 0, 8)$.

In the second pattern, T and the seven pure odd heptads supply eight pairwise disjoint Lagrangians: each pure odd heptad is $z + (L \setminus \{0\})$. Lemma 6.2 identifies the remaining odd projections as $M \setminus \{0\}$. These are exactly the projections of the seven mixed odd members. Every even point $u \in M \setminus \{0\}$ is orthogonal to every such odd vector $z + e$, so it cannot use a mixed label. It cannot use the label containing z . Nor can it share a pure odd label: a linear functional on its Lagrangian cannot take value one at every nonzero point, since its values on $e, f, e + f$ sum to zero. Only the three pure even labels remain. But the seven nonzero points of M form a clique and need seven labels, a contradiction.

Both patterns are impossible, so $\chi(\Gamma_7) \geq 19$. The induced-subgraph inclusion $\Gamma_7 \subseteq \mathcal{O}_7^*$ gives the full-graph inequality. The clique formula supplies the final comparison with 16. $\square$

This proof leaves the exact chromatic numbers open. It needs neither a nineteen-color witness nor the further profile exclusions below. The quadratic and incidence calculations have exact coordinate checks described in Section 8.

7 Nineteen-color constraints and higher-dimensional questions

Theorem 6.3 proves a numerical lower bound; deciding whether nineteen colors suffice is a separate problem. We give common necessary conditions for a hypothetical nineteen-color line coloring and a general reduction for the full-subspace problem.

7.1 Deficits, projected charges and hole capacity

Let C_z be the class containing z , and put $s = |C_z|$. Every other member of C_z is odd, and Lemma 6.1 gives $s \leq 8$. The other eighteen classes have capacity seven and contain $127 - s$ vertices. If $D_1, \dots, D_r$ are the undersized classes, with sizes k_i , then

$$\sum_{i=1}^r (7 - k_i) = s - 1. \quad (5)$$

Write m_i, o_i for their even and odd counts, $m = \sum_i m_i$, and let A, B, C count the saturated pure even, mixed and pure odd heptads. Counting both parities gives

$$7A + 6B + m = 63, \quad B + 7C + \sum_i o_i + s = 64, \quad A + B + C = 18 - r.$$

Using $\sum_i k_i = 7r - (s - 1)$, these equations become

$$B = m + 7t, \quad A = 9 - m - 6t, \quad C = 9 - r - t \quad (6)$$

for an integer t with nonnegative counts. The same formulas apply to every characteristic size; s determines the defect patterns through (5).

Let $\pi(v) = v + (v \cdot z)z$, $h = \sum_{v \in C_z} \pi(v)$, and $w_i = \sum_{v \in D_i} \pi(v)$. Every saturated class has projected sum zero. For a pure even heptad this was proved in Theorem 6.3; for a pure odd heptad it follows by summing a Lagrangian's nonzero points. In a mixed class the six even vectors form a basis of E , since $J_6 + I_6$ is invertible. Their sum and the odd member's projection both pair to one with each basis member and are therefore equal. The full projected sum is zero, so

$$\sum_i w_i = h. \quad (7)$$

For a quadratic refinement q of E , an independent class with k members and o odd members satisfies

$$\sum_v q(\pi(v)) = q\left(\sum_v \pi(v)\right) + \binom{k}{2} + \binom{o}{2} \quad \text{in } \mathbb{F}_2.$$

The characteristic projections are pairwise orthogonal, and contribute $q(h)$. A pure even or mixed heptad contributes one; a pure odd heptad contributes zero. Summing over the partition and polarizing (7) gives

$$\sum_{i < j} w_i \cdot w_j = A + B + \sum_i \left(\binom{k_i}{2} + \binom{o_i}{2} \right) \quad \text{in } \mathbb{F}_2. \quad (8)$$

These identities allow $h = 0$ and do not assume that individual defect charges lie in a hole Lagrangian.

There is a common capacity inequality whenever the C pure odd heptads leave $d = 9 - C$ disjoint hole Lagrangians. Every pure even heptad meets each isotropic side at most once. A mixed class's odd projection forbids one whole side, so its even members occupy at most $d - 1$ holes. A pure odd defect contributes no even hole point; a pure even defect contributes at most d ; a defect with both parities contributes at most $d - 1$. Thus

$$7d \leq dA + (d - 1)B + \sum_i b_d(k_i, m_i), \quad b_d(k, m) = \begin{cases} 0, & m = 0, \\ d, & m = k, \\ d - 1, & 0 < m < k. \end{cases} \quad (9)$$

For $C = 8$ the required completion is Lemma 6.2. For $C = 6, 7$ it uses the finite completion results in the accompanying package [11]: six or seven disjoint Lagrangians leave respectively three or two uniquely determined hole Lagrangians. Their proof consists of a written symplectic normalization and two independent exhaustive enumerations; these finite premises are separate from the written proof of Theorem 6.3.

For $s = 4$, the outside deficit is three, giving patterns 4, 5 + 6 and 6 + 6 + 6. Their parity-count catalogs contain 9, 46 and 48 rows. The accompanying further analysis excludes the single-four pattern, nineteen five-plus-six rows and nine three-six rows, leaving 27 and 39 count candidates. They are necessary profiles, not realized partitions. Of the remaining 66, 64 have $C \in \{6, 7, 8\}$ and already satisfy (9); two have $C = 5$. The profile $(m_5, m_6; A, B, C) = (5, 4; 6, 2, 8)$ has four necessary marked forms; three are excluded by independently checked covering certificates, leaving the zero-five-even-sum form. In the most recent exclusion, a written pointwise-hole symplectic group of order 64 reduces 127984 necessary remaining sets to 2306 representatives. A finite failed-state proof excludes all representatives, hence the mixed characteristic-charge form. This is a partial case analysis. The package records its written reductions and finite dependencies; it supplies no uniform exclusion of all remaining profiles.

The further analysis also excludes $s < 4$. Characteristic sizes 4 through 8 remain open, and larger s introduce outside deficits 4 through 7. Closing the 66 size-four count candidates would therefore not finish the nineteen-color problem. Excluding every nineteen-color line coloring would improve the lower bound to twenty, without establishing a matching upper bound. Conversely, a nineteen-color line witness would still require an extension check before it proved a full-subspace upper bound.

7.2 A dimension reduction

Proposition 7.1. *Let K be a fixed clique in $\mathcal{O}_n^*$ of size at most k , precolored with distinct labels from k colors. If $N(n - d) < k$, every uncolored d -dimensional vertex can be deleted by the degree-versus-list rule. The residual instance is k -colorable if and only if the full graph is k -colorable.*

Proof. Every neighbor of a d -space U is a nonzero subspace of $U^\perp$, so $\deg U \leq N(n - d)$. If f fixed clique vertices are adjacent to U , its residual degree is at most $N(n - d) - f$ and its list has size $k - f$. A color is therefore available when U is restored. Apply this argument in reverse deletion order. $\square$

For $n = 7$, the nonzero subspaces of $T = \text{span}(3, 12, 48)$ together with $\text{span}(64)$ give a 16-clique. Since $N(3) = 15$, vertices of dimension at least four can be deleted in any coloring problem with at least sixteen labels and these fixed clique colors. The lower bound in Section 6 rules out sixteen,

seventeen and eighteen colors; the reduction remains available for the nineteen-color extension problem.

The exact chromatic numbers for $n \geq 7$ and the question of infinitely many dimensions with $\chi > \omega$ remain open. Within dimension six, a geometric construction of an optimal full-subspace coloring is a separate problem. The negative results in Section 5 show that such a construction must use enough placement information to break the stated symmetry.

Problem 7.2. Can one specify auxiliary geometric data and a nontrivial proper subgroup $H < \text{Stab}_{O_6(\mathbb{F}_2)}(T)$, and construct a proper 15-coloring c from those data with $c(hU) = h \cdot c(U)$ for $h \in H$, where labels are the nonzero subspaces of T ? The construction should prove availability and separation without looking up the archived coloring certificate.

The checked twelve-coloring already determines $\chi(\Gamma_6)$; passing to all nonzero subspaces increases the chromatic number by exactly three.

8 Computational certificates and formal verification

The positive coloring certificates are checked using integer arithmetic and the coordinate definition of orthogonality. The full checker imports neither the search program nor a SAT solver. It establishes enumeration completeness from distinctness and Gaussian counts, then checks every pair of vertices. Three deliberately invalid certificates, respectively omitting a vertex, duplicating a subspace, and identifying the colors of adjacent coordinate lines, are rejected. The line-coloring checker independently verifies its partition and all within-class pairs. For the twelve-color certificate, `develop/check_line_chromatic_exact.py` checks all 1953 distinct vertex pairs, including all 961 edges. It independently reproduces every clause of the normalized search encoding and its SHA-256 digest, verifies the assignment against those clauses, and rejects duplicate-vertex, zero-vector and monochromatic-edge controls.

The radical/quotient clique theorem has an existing Lean formalization. Its source is `formal/NikandishClique.lean`. The theorem is `NikandishClique.result`, inside the namespace

`D5.S3.Combinatorics.Orthogonality.`

The upstream revision and source hash are recorded in `formal/UPSTREAM.json`; dependencies must be obtained from that pinned upstream checkout.

The six-dimensional development uses vectors in $\text{Fin}(64)$, XOR addition, and the coordinate dot product modulo two. A vertex is a zero-containing, XOR-closed subset other than $\{0\}$, exactly the nonzero binary subspaces in this representation. The formal graph is simple and uses coordinate orthogonality between distinct such subsets. The catalogue contains 2825 entries including zero and 180800 adjoining-vector witnesses. The theorem `Catalogue.complete` derives coverage from the zero entry and closure under adjoining any vector; it does not assume the Gaussian enumeration counts. The theorem `Catalogue.colorable` transfers the checked catalogue coloring to this graph. A checked 15-clique supplies the converse inequality in `BinaryOrthogonality.chromatic_number_six`.

The archived Lean 4.33.0 development verifies four finite assertions with `native_decide`. The final theorem depends on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, together with four native-evaluation axioms; its computational trust therefore includes native evaluation. The generic completeness and colorability lemmas depend only on the three standard axioms. The recovered original log corrects the incomplete historical summary; see `notes/formal-verification-audit.md`. The exhaustive integer certificate independently verifies the

same coloring. The weighted line bound, twelve-color line certificate, geometric profile counts and dimension-seven reduction are not covered by this Lean theorem.

The public package [11] contains the literal witnesses, independent checkers, search records, and SHA-256 bindings. The new geometric profiler checks every assigned color against its $H(U)$ -determined list. The dimension-seven reduction is also replayed in dimension six, where it reproduces the established core size and Boolean encoding counts.

The dimension-seven lower bound in Section 6 has a written proof. The checker `develop/check_dimension_seven_nineteen_obstruction.py` independently verifies its coordinate ingredients: all 4096 quadratic polarization pairs, the 36/28 quadratic value counts, all 288 even heptads and 2016 six-point cliques with their unique heptad completions, and all 8505 point–hyperplane intersections with the 135 Lagrangians. Nine explicit spread controls check the hole-clique obstruction. These controls do not enumerate all eight-member partial spreads; Lemma 6.2 proves their completion uniformly.

The additional nineteen-color constraints in Section 7 have their own evidence. The six- and seven-member partial-spread completion lemmas use written normalization and exhaustive finite enumerations. The three marked-form exclusions use independently checked covering certificates. For the mixed characteristic-charge form, `develop/check_dimension_seven_five_four_G_cover.py` reconstructs the complete necessary-root family and orbit partition. It then verifies 3351 reachable failed states, all 1045 branches and 2401 leaves, using all 224 anchored heptads. Every branch removes seven points, leaves admit no row, and the empty set is never declared failed; induction proves the exclusion. Three invalid certificates are rejected. The proof notes, checkers and hash-bound reports are recorded separately in the public package. The small accounting auditor `develop/check_dimension_seven_profile_scope.py` reconstructs the parity-count catalogs and checks the scope of the remaining count candidates, including the 64 rows to which the common hole capacity applies. Its archived marked-form scope predates the latest G exclusion; the G covering auditor verifies that update separately. Neither check excludes the remaining zero-five-even-sum form. None of these dimension-seven results is covered by the archived six-dimensional Lean theorem.

Declaration of generative AI use

During preparation of this manuscript, DeepSeek (through deepseek.com and its mobile application) was used for code generation, diagnostic analysis, and language editing. OpenAI’s Codex was used for mathematical exploration and proof development, code generation, formalization support, verification diagnostics, and manuscript editing. Mathematical arguments and computational certificates are supported by the proofs and checks described in this paper; the scope of formal verification, including native evaluation, is stated in Section 8. AI-generated material requires author review. The authors take full responsibility for the mathematical claims, code, certificates, and final text. These tools are not authors and bear no responsibility for the content.

References

- [1] D. D. Anderson and M. Naseer, *Beck’s coloring of a commutative ring*, J. Algebra **159** (1993), 500–514.
- [2] D. F. Anderson and P. S. Livingston, *The zero-divisor graph of a commutative ring*, J. Algebra **217** (1999), 434–447.
- [3] I. Beck, *Coloring of commutative rings*, J. Algebra **116** (1988), 208–226.

- [4] M. Behboodi and Z. Rakeei, *The annihilating-ideal graph of commutative rings I*, J. Algebra Appl. **10** (2011), 727–739.
- [5] M. Behboodi and Z. Rakeei, *The annihilating-ideal graph of commutative rings II*, J. Algebra Appl. **10** (2011), 741–753.
- [6] A. E. Brouwer, A. M. Cohen and A. Neumaier, *Distance-Regular Graphs*, Springer, 1989.
- [7] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Springer Universitext, 2012.
- [8] P. J. Cameron, *Projective and Polar Spaces*, QMW Maths Notes **13**, Queen Mary and Westfield College, 1991.
- [9] C. Godsil and G. Royle, *Algebraic Graph Theory*, Springer GTM **207**, 2001.
- [10] L. Lovász, “Kneser’s conjecture, chromatic number, and homotopy,” J. Combin. Theory Ser. A **25** (1978), no. 3, 319–324.
- [11] H. Ma, R. Nikandish and W. Zhang, *Binary subspace orthogonality graphs: research and reproducibility workspace*,
<https://github.com/the-omega-institute/binary-subspace-orthogonality>.
- [12] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977.
- [13] R. Nikandish, *Annihilating-Ideal Graphs and Orthogonality Graphs over $\mathbb{F}_2$* , arXiv:2609.22769v1, 2026.